

# Nazmath Pasha

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GitHub | LinkedIn | Portfolio

## Profile Summary

Electronics and Communication Engineering student with hands-on project experience in **Embedded Systems and IoT development**. Proficient in STM32, ESP32, Arduino, and Embedded C/C++. Experienced in sensor interfacing, communication protocols, and basic debugging techniques. Interested in building reliable embedded solutions and continuously improving technical skills. Collaborated in team-based projects, contributing to design, implementation, and system testing. Quick learner with a strong interest in understanding low-level system behavior and problem-solving.

## Education

**Bachelor of Engineering – Electronics and Communication Engineering** 2022 – 2026

SJC Institute of Technology, Karnataka

CGPA: 8.5 / 10

**Pre-University Course (PUC) – Science** 2020 – 2022

SBGNS Rural Composite PU College, Chikkaballapur

Percentage: **87.67%**

## Technical Skills

**Embedded Platforms:** STM32, ESP32, Raspberry Pi, ARM Cortex-M

**Programming:** Embedded C, C++, Python

**Microcontroller Peripherals:** GPIO, ADC, PWM, Timers, Interrupts

**Communication Protocols:** UART, SPI, I<sup>2</sup>C, MQTT

**Tools & IDEs:** STM32CubeIDE, Keil uVision, VS Code, Arduino IDE, MATLAB, Multisim, Git, Linux, ThingSpeak, Blynk, Wokwi

**Debugging Tools:** Oscilloscope, Logic Analyzer, Serial Debugging, Log Analysis

**Concepts:** Firmware Development, Sensor Interfacing, RTOS Basics, Hardware-Software Integration, Embedded Linux

## Embedded & IoT Projects

**IoT-Based Women's Safety Patrolling Robot** 2025

*ESP32, Embedded C, Blynk IoT*

- Developed control logic for autonomous patrolling and monitoring.
- Integrated sensors and communication modules for real-time data acquisition and alert generation.
- Performed hardware-software debugging using log analysis and systematic testing.
- Documented testing procedures and troubleshooting steps.

**Women Safety Smart Watch (IoT Wearable Device)** 2025

*ESP32, Embedded C, GPS*

- Designed and implemented an IoT safety device capable of sending emergency alerts with GPS location.
- Interfaced GPS and GSM modules using UART for real-time tracking and SMS alerts.
- Implemented device control logic for emergency activation and alert handling.
- Tested communication reliability under varying network conditions.

**Smart Mushroom Growth Chamber with Environmental Automation** 2025

*ESP32, Sensors, Embedded C*

- Developed automated environmental control for temperature and humidity regulation.
- Interfaced sensors and actuators using GPIO and I<sup>2</sup>C.
- Implemented threshold-based control logic for environmental automation.
- Performed functional testing to validate system behavior.

**Automated Toll Collection System** 2024

*Arduino, Embedded C*

- Developed firmware for sensor interfacing and barrier control logic.
- Implemented UART/SPI communication between modules.
- Used timers and interrupts for precise barrier operation and vehicle detection.
- Implemented debounce handling to reduce false triggering from sensor noise.
- Conducted repeated testing to verify system reliability.

## Experience

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### Intern – Talent Trek

Apr 2025 – Sep 2025

- Worked on RTL debugging and verification, strengthening systematic debugging skills.
- Assisted in simulation, issue reproduction, and documentation of technical findings.
- Applied structured validation practices relevant to embedded firmware testing.
- Performed waveform analysis and signal tracing during simulation to identify logic mismatches.
- Created detailed bug reports and validation notes, improving issue tracking and resolution time.
- Gained hands-on exposure to debugging methodology applicable to embedded firmware validation.

## Achievements

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- Selected for **Smart India Hackathon (Institute Level)**.
- Mentored **20+ students** in embedded mini-project development and debugging.
- Participated in national-level hackathons: Aavishkar (SJCIT), Kagada (UVCE), Tech Utsav (CMR University).
- Built and demonstrated multiple embedded prototypes for academic and hackathon evaluations.
- Actively contributed to peer learning by guiding classmates in sensor interfacing and firmware logic.
- Completed 30+ hands-on experiments involving microcontrollers, sensors, and communication modules.

## Certifications

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- Talent Trek Internship Completion Certificate
- Edunet Foundation (AICTE & IBM SkillsBuild) – Cloud and IoT
- NIELIT – Embedded Systems, Python
- Coursera – Embedded Systems Software and Development Environments
- Cisco Networking Academy – Python Essentials, Introduction to IoT
- SkillUp – Embedded Systems, IoT
- TCS iON – Communication Skills